Centered Difference Interpolation - Stirling and Bessel Methods

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2. Setup for Centered Interpolation

1. Choose central point(s): x_0 (Stirling), or x_0, x_1 (Bessel). 2. Label points symmetrically:

$$x_{-m}, \dots, x_{-1}, x_0, x_1, \dots, x_m$$

3. Compute **divided differences table** up to required order. 4. Define normalized distance:

$$s = \frac{x - x_0}{h}, \quad h = \text{uniform step size}$$

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3. Stirling's Method (Odd Points)

Formula (as per Maqsood Alam, video):

$$\begin{aligned} P_n(x) = & f[x_0] + s\frac{\Delta y_0 + \Delta y_{-1}}{2} + s^2\Delta^2 y_{-1} + \frac{s(s^2-1)}{2!}(\Delta^3 y_{-1} + \Delta^3 y_{-2}) \\ & + \frac{s^2(s^2-1)}{4!}\Delta^4 y_{-2} + \dots \end{aligned}$$

Key points:

- Odd-order differences $\rightarrow$ average of two symmetric differences.
- Even-order differences $\rightarrow$ central differences.
- Highest-order term requires exact central anchor.
- Even number of points $\rightarrow$ drop last term (no central anchor).

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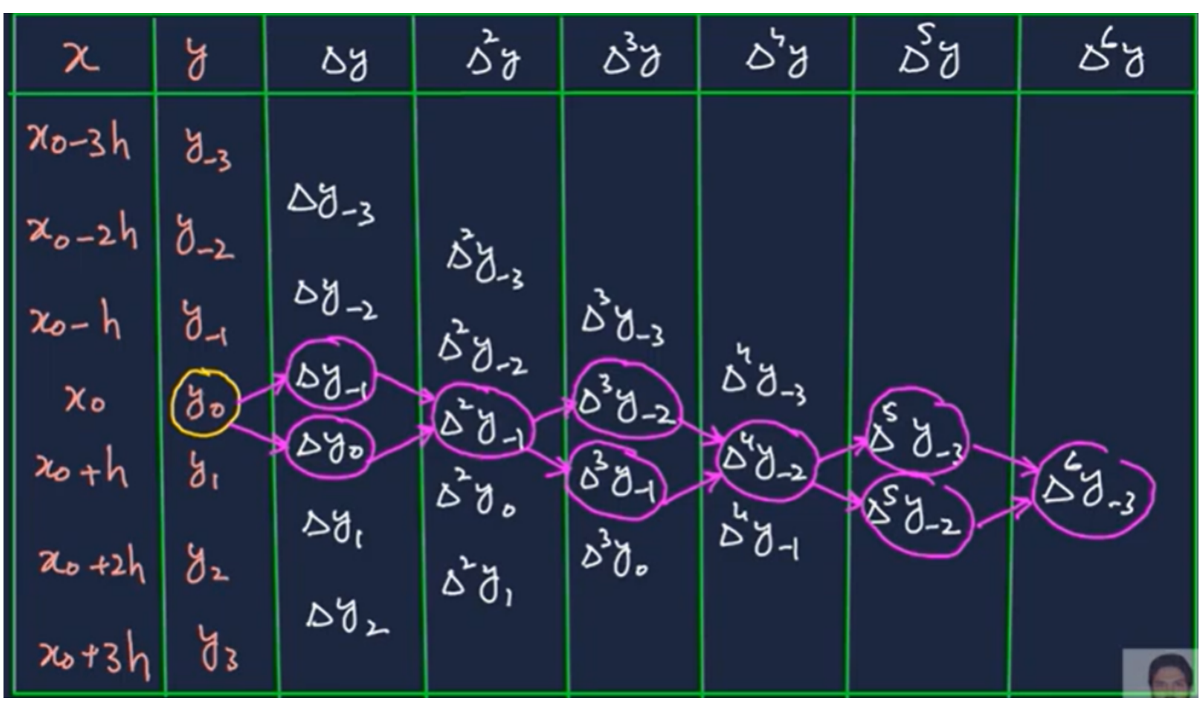
4. Stirling Difference Table (Visualization)



5. Stirling Example (5 points)

x	$f(x)$	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1.0	0.7651977	-0.1451117	-0.0036447	0.0041234	-0.000218
1.3	0.6200860	-0.164684	0.002134	-0.002345	
1.6	0.4554022	-0.173584			
1.9	0.2818186	-0.171284			
2.2	0.1103623				

- Central point: $x_0 = 1.6$, step $h = 0.3$, compute $s = \frac{x-x_0}{h}$ - Apply Stirling formula term by term using the table.

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6. Bessel's Method (Even Points)

Used when: - Even number of points, interpolation between two central points x_0, x_1.

Formula (as per Maqsood Alam, video):

$$f(x) \approx \frac{f_0 + f_1}{2} + s\Delta f_0 + \frac{s^2}{2!}\Delta^2 f_0 + \frac{s(s^2-1)}{3!}(\text{average of 3rd differences}) + \dots$$

Key points:

- First term = average of two central points.
- Odd/even differences handled as in Stirling.
- Symmetry preserved, all points used.

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7. Bessel Example (4 points)

x	$f(x)$	Δy	$\Delta^2 y$	$\Delta^3 y$
1.3	0.6200860	-0.165684	-0.024110	
1.6	0.4554022	-0.173584	0.014123	
1.9	0.2818186	-0.171284		
2.2	0.1103623			

- Interpolation point: $x = 1.75$ - Central points: $x_0 = 1.6$, $x_1 = 1.9$ - Step: $h = 0.3$, $s = \frac{1.75-1.6}{0.3} = 0.5$ - First term: $(f_0 + f_1)/2 = 0.3686104$ - Apply Bessel formula term by term.

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8. Comparison: Stirling vs Bessel

- Stirling: odd points, exact central anchor, last term dropped if even points.
- Bessel: even points, average of central pair as first term, symmetry preserved.
- Both: odd differences averaged, even differences central.
- Both minimize error near table center.

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